

Diagnose Before Simplifying: Operation-Aware Biomedical Text Simplification

Sruthilaya Umasankari Soma Shanmuga Sundaram (umasankarisomashan.s@northeastern.edu), and
Sophakotra Son (son.so@northeastern.edu), and
Zihao Wang (wang.zihao10@northeastern.edu), and
Rishabh Gowda (gowda.ri@northeastern.edu)
Northeastern University

Abstract

Biomedical text simplification is often treated as a single generation task, even though different sentences require different types of edits and some content must be preserved exactly. We propose an operation-aware simplification framework that first predicts whether a sentence requires Substitution, Explanation, or Generalization, then applies an operation-specific strategy rather than using unrestricted generation uniformly. The system combines multi-signal complexity detection, UMLS-based jargon identification, a weakly supervised operation classifier, deterministic CHV substitution, and constrained FLAN-T5 generation. BioBERT achieves the strongest operation-classification performance among the evaluated models (0.465 macro-F1 vs. 0.424 for TF-IDF and 0.392 for DistilBERT), while engineered UMLS features further improve the lexical baseline. We additionally introduce a Detect-Twice, Verify-or-Fallback mechanism for the Substitution path, where detected numerical values and warning phrases are reused as post-generation constraints and outputs that violate them are rejected. On the held-out PLABA test set, the operation-aware pipeline improves SARI over rule-based CHV (21.72 vs. 21.12) while maintaining perfect warning preservation, and preserves warnings and numerical content better than direct FLAN-T5 generation. On warning-bearing sentences, it also exceeds direct generation in SARI (25.43 vs. 24.73), warning preservation (1.000 vs. 0.980), and numerical preservation (0.909 vs. 0.818), though this subset is small ($n = 49$). Finally, we show that random evaluation can miss rare safety-critical failures: a random 50-sentence sample showed perfect warning preservation for all systems, while warning-stratified evaluation exposed a failure in unguided generation. Together, these results support operation-aware routing as a controlled middle ground between lexical substitution and unconstrained generation, and motivate targeted stress testing for safety-sensitive simplification.

1 Introduction

Patient-facing biomedical text – research abstracts, clinical notes, drug labeling – is frequently written at a reading and vocabulary level that assumes clinical training the intended reader does not have. Low health literacy is linked to worse health outcomes (Berkman et al., 2011), so making biomedical text readable is a practical clinical concern, not only a linguistic one. Automated simplification systems aim to close this gap, but most treat simplification as a single, undifferentiated transformation applied uniformly across a text: a sentence is fed into a model and a simpler sentence is returned, whether the difficulty came from a single jargon term, a tangled clause structure, or a critical dosage.

We argue this is the wrong framing. PLABA’s own five-operation taxonomy (Attal et al., 2023) shows that “simplify” is not one transformation but several distinct ones – Substitution, Explanation, Generalization, Omission, and Exemplification – and prior work at the TSAR 2025 shared task confirms that no submitted system, across 48 submissions from 20 teams, classifies which operation a span needs before generating output (Alva-Manchego et al., 2025). We take a diagnostic approach instead: classify first, then simplify. In this paper, we operate at the *sentence* level, not the sub-sentence span level the general problem is sometimes framed at in prior work – classifying one operation per source sentence, rather than identifying and classifying multiple spans within a single sentence, is the scope of our pseudo-labeling and classifier (Section 3.3). Before any text is rewritten, we identify what type of complexity a sentence carries, and only then apply the operation suited to it. A jargon term can be replaced from a controlled vocabulary with no free-form generation involved, avoiding generative hallucination at this step. An unfamiliar concept can be explained. An overly specific statement can be generalized.

Routing each sentence to the right operation lets us simplify aggressively where it is safe and stay conservative where information must be preserved.

Our contributions are: (1) a methodological finding, independent of any specific system, that random sampling can miss rare but safety-critical failures, motivating stratified stress testing alongside aggregate evaluation (Section 3.1 demonstrates this directly: a random sample and a stratified sample of the same systems yield qualitatively different safety conclusions); (2) operation-aware sentence routing supported by a multi-signal complexity detection framework (five complementary detectors, near-zero pairwise correlation among the four with non-degenerate hit rates, strongest $\phi = 0.078$) and a BioBERT operation classifier (TF-IDF $\rightarrow$ DistilBERT $\rightarrow$ BioBERT progression, plus a controlled ablation of which detected features help classification versus which are better reserved for generation-time safety constraints); (3) an end-to-end evaluation of an operation-aware biomedical simplification pipeline against uniform baselines (rule-based CHV and direct FLAN-T5), showing it improves readability over safe-but-limited rule-based substitution while retaining that baseline’s safety profile; and (4) our **Detect-Twice, Verify-or-Fallback** mechanism (Section 2), currently implemented for our Substitution-routed generation path (Section 3.4 discusses why this verification does not yet extend to our Explanation/Generalization generation path).

PLABA annotations show that simplification involves distinct operations: Substitution (64.7%), Explanation (19.3%), Generalization (6.3%), with the remainder split between Omission and Exemplification (Ondov et al., 2026). These operations are not interchangeable: lexical lookup cannot produce explanations, while unrestricted generation can modify information that should remain fixed. This asymmetry is especially important in biomedical text. For example, simplifying “*Do not exceed 2.5 mg daily*” to “*Take the medicine daily*” improves surface readability while deleting both a warning and an exact dosage. We therefore evaluate readability jointly with warning and numerical preservation (Section 3.1).

2 Main Idea

Figure-free summary of the architecture: a source sentence is first passed through five complexity detectors (Section 2.1) for corpus-level diagnosis and

safety-constraint extraction, and separately through an operation classifier that predicts one of Substitution, Explanation, or Generalization (Section 3.3).

The predicted operation determines the generation strategy: Substitution routes to a Consumer Health Vocabulary (CHV) lookup, a deterministic, auditable dictionary substitution with substantially lower hallucination risk than free-form generation; Explanation and Generalization route to a constrained large language model (FLAN-T5) prompted with an operation-specific instruction, rather than a single generic “simplify this” prompt.

A single pattern underlies how we use complexity detectors twice, not once: we call it **Detect-Twice, Verify-or-Fallback**. The same complexity detectors evaluated as routing signals are reused downstream as explicit, checkable generation constraints: after generation, their output becomes a condition verified post-hoc, not a soft instruction the model is merely asked to follow. If the requirement is violated, the pipeline rejects the generation outright and falls back to a deterministic alternative, rather than accepting whatever the model produced. Section 3.4 instantiates this concretely for our Substitution-routed generation step. Which detector outputs feed the classifier as features, and how reliable our pseudo-labels are, are both empirical questions we address with results rather than assumptions, in Section 3.3.

2.1 Complexity Detection Framework

Before a biomedical sentence can be simplified, the system must first determine *what kind* of complexity it exhibits. We argue that biomedical text complexity is not a single dimension but a composite of at least five distinct signals, each requiring a different detection paradigm:

- **Numerical expressions** (dosages, risk ratios, *p*-values, confidence intervals), detected via regular expressions. These patterns are finite and rule-governed, so no learned model is needed to achieve full recall on known formats.
- **Safety-critical warning language** (contraindications, adverse effects, risk alerts), detected via a hand-built lexicon, negation-aware (a sentence stating a warning does *not* apply is not flagged). Because these phrases are clinically standardized, a lexicon provides an auditable, deterministic warning signal without requiring generative inference.

- **Named biomedical entities** (diseases, chemicals), detected via a supervised NER model (SciSpaCy, BC5CDR-trained; [Neumann et al., 2019](#)). General-domain NER models miss domain-specific entities such as *nephrotoxicity*; biomedical pretraining is not optional for this signal.
- **Structural complexity**, detected via dependency parse depth rather than raw sentence length. Length alone unfairly penalizes long-but-simple sentences (e.g., coordinated lists of facts), whereas parse depth captures genuine syntactic nesting that a vocabulary-only view misses entirely.
- **Medical jargon**, detected via UMLS concept matching (Section 2.2). A term is treated as jargon if it exists in UMLS as a formal concept but is not part of common English vocabulary.

No single paradigm captures all five signals: regular expressions catch numbers but miss entities; a lexicon catches warnings but misses syntactic structure; NER catches entities but misses jargon that is not a named entity in the BC5CDR sense (e.g., procedure names, qualifiers). We treat this as the central architectural argument motivating a multi-detector design rather than a single end-to-end classifier – an argument reinforced empirically in Section 3.2, where pairwise correlation among the detectors is near zero wherever it is statistically defined.

2.2 QuickUMLS Integration

Our jargon detector is built on the Unified Medical Language System (UMLS), a meta-thesaurus unifying over 200 source vocabularies (SNOMED CT, MeSH, RxNorm, and others) into a single concept space of several million entries. We use QuickUMLS ([Soldaini and Goharian, 2016](#)), which maps free text to UMLS concept identifiers via approximate string matching, to perform this lookup at the sentence level.

Definition of jargon. A term is flagged as jargon if it resolves to a UMLS concept *and* is not part of common English vocabulary, operationalized via three filters layered on top of raw QuickUMLS output:

1. **Semantic type restriction.** UMLS concepts are tagged with one or more of 127+ semantic

types. We restrict matches to eight clinically relevant types (e.g., *Disease or Syndrome*, *Pharmacologic Substance*, *Therapeutic or Preventive Procedure*), excluding generic administrative or qualifier concepts that are technically present in UMLS but are already plain English (e.g., *primary*, *performed*).

2. **Similarity threshold.** We require a QuickUMLS similarity score of at least 0.85, filtering out loose or uncertain string matches.
3. **Lexical frequency filter.** Semantic-type restriction alone is insufficient: some common English words coincidentally match real, clinically-relevant UMLS concepts. For example, *control* matches a genuine pharmacologic-substance concept at similarity 1.0 – the same semantic type needed to correctly catch real drug names such as *sorafenib*. We therefore apply a word-frequency filter using Zipf frequency scores ([Speer, 2022](#)), requiring single-word matches to fall below a threshold of 3.8 to be treated as jargon. This threshold was validated against a held-out set of 16 known common and jargon terms, cleanly separating common words (*control*: 5.40, *determine*: 4.63, *disease*: 4.90) from genuine jargon (*carcinoma*: 3.15, *hyperglycemia*: 2.31, *sorafenib*: 1.43). A preliminary attempt using a broad English wordlist (NLTK’s words corpus) was rejected: it classified genuine jargon such as *hyperglycemia* as ordinary English, since raw dictionary membership does not capture actual usage frequency.

Additionally, since QuickUMLS returns every candidate concept considered for a matched span rather than a single best match, we deduplicate by retaining only the highest-similarity candidate per span.

3 Details, Evaluation, and Results

3.1 Safety Evaluation Methodology

We measure whether a simplification system preserves safety-critical content (contraindications, adverse effect warnings) using a warning-preservation metric with paraphrase matching, rather than surface *n*-gram overlap metrics such as ROUGE. ROUGE cannot distinguish between a system that preserves a warning and one that silently drops it,

provided the rest of the sentence is scored favorably; our metric targets exactly this failure mode. We similarly measure numerical preservation (exact dosage and statistic survival) and, where a NER environment is available, entity preservation.

A central methodological finding from our evaluation is that **random sampling can be statistically underpowered to detect rare safety failures**. Warning-bearing sentences occur at an overall rate of approximately 5.7% across the combined train, validation, and test sentence-level pairs before source-level deduplication (519/9,085). The corresponding held-out split rates differ slightly: 6.4% on validation (73/1,141) and 4.9% on test (49/1,000). Separately, 31.3% of training abstracts contain at least one warning-bearing sentence; this is an abstract-level statistic and therefore not directly comparable to the sentence-level rates above. A random sample of 50 sentences showed all baseline systems achieving perfect warning preservation, a misleadingly optimistic result. Only a *stratified* stress test – restricted to warning-bearing sentences specifically – revealed that unguided LLM-based simplification is the only baseline that drops safety-critical content under evaluation.

The warning-stratified samples are small ($n = 30\text{--}73$), so we interpret preservation differences cautiously; for example, 0.980 on the 49-sentence test subset may represent approximately one failure. Our claim is therefore methodological rather than a broad claim that generative systems are unsafe: stratified evaluation exposed a failure that a random 50-sentence evaluation missed entirely. This motivates targeted stress testing alongside aggregate metrics for safety-critical NLP.

3.2 Complexity Coverage Results

Table 1 reports detector flag rates across our verified training split ($n = 530$ abstracts¹), comparing our real UMLS-based jargon detector against a morphological (suffix-pattern) fallback baseline.

UMLS detected at least one jargon candidate in 100.0% of training abstracts, compared with 86.8% under the morphological fallback. We are careful

¹We note a discrepancy between this figure and an earlier internal estimate of $n = 635$; the current figure was independently re-derived via two methods (direct exclusion of validation/test PMIDs from the source data, and the shipped train/validation/test split file) which agree with each other. train.csv contains 635 rows but 531 unique PMIDs (530 after excluding one row with a blank PMID) – 104 PMIDs have two adaptation-version rows each. We report the verified, deduplicated number here.

Detector	Morph.	UMLS
NER	–	98.7%
Warning cues	–	31.3%
Syntactic depth	94.0%	94.0%
Numerical exprs.	–	60.4%
Jargon (UMLS)	86.8%	100.0%
3+ detectors firing	90.8%	98.1%

Table 1: Percentage of abstracts ($n = 530$) in which each detector fires at least once – an abstract-level hit-rate, not a per-mention recall figure. “Morph.” denotes a suffix/pattern heuristic fallback used when a full UMLS installation is unavailable; “UMLS” denotes the full QuickUMLS pipeline described in Section 2.2.

about what this figure does and does not claim: it is an abstract-level hit-rate – whether *any* jargon was flagged per abstract – not a per-mention recall measurement, and it does not establish that every individual jargon expression in the dataset was detected. Read this way, it is still a meaningful, direct demonstration of a coverage gap between rule-based jargon detection and true terminology-base lookup: the morphological heuristic fails to flag any jargon candidate in 13.2% of abstracts where the UMLS pipeline still fires. We validated this result in two stages: first against a small set of targeted test sentences containing known jargon (e.g., *hepatocellular carcinoma*, *sorafenib*, *nephrotoxicity*), and second via manual review of a random sample of ten training abstracts, confirming that flagged terms were genuine clinical terminology (e.g., *implantable cardioverter-defibrillator*, *hyporeninemic hypoaldosteronism*, *gabapentinoids*) rather than residual false positives.

Consistent with our preliminary findings, the majority of abstracts exhibit multiple simultaneous complexity signals: 98.1% of abstracts trigger three or more of our five detectors under the full UMLS pipeline. Note that individual detector hit-rates vary widely (Table 1): UMLS jargon and NER fire on nearly every abstract (100.0% and 98.7% respectively), while warning cues and numerical expressions fire on a minority (31.3% and 60.4%). At the abstract level, UMLS jargon fires on all 530 abstracts and therefore has zero variance, making its phi correlation undefined. Among the remaining four non-degenerate detectors, pairwise correlations are near zero (strongest $\phi = 0.078$), indicating complementary signals. We therefore do not claim five detectors are optimal, only that the evaluated signals capture distinct dimensions

of biomedical complexity.

3.3 Operation Classification

Predicting the operation a sentence requires is the research centerpiece. We classify at the sentence level – one operation label per source sentence – rather than identifying and labeling operations for sub-sentence spans; extending to finer-grained span-level classification within a sentence is future work (Section 5). Crucially, the classifier sees only the *source* sentence: at inference time no target exists, so any feature derived from the reference would be leakage.

Pseudo-labeling. Because PLABA lacks public sentence-level operation labels, we derive them by weak supervision from aligned source-target pairs, sentence-aligning data. json and excluding all validation and test PMIDs to prevent leakage. Each pair is labeled by a length-ratio rule with an edit-distance tiebreak: a target at least 15% longer is an Explanation, one at least 15% shorter is a Generalization, and a similar-length pair with sufficient character-level change is a Substitution. This yields 6,307 labeled training pairs: 43.2% Substitution, 34.4% Explanation, 22.5% Generalization.

The pseudo-label distribution (43.2/34.4/22.5%) differs substantially from human operation frequencies (64.7/19.3/6.3%; Ondov et al., 2026), especially over-representing Generalization by roughly $3.6\times$. Length-based weak supervision can confuse shortening caused by omission with true generalization, so these labels are not treated as gold. We therefore interpret classification as a proof of concept; this task is itself difficult, with Ondov et al. (2026) reporting human inter-annotator F1 of 0.46.

Classifier progression. Table 2 reports macro-F1 across three classifier backbones: a lexical baseline, a general-domain contextual model, and a biomedical-domain contextual model. Classification is evaluated per aligned pair, whereas SARI (Section 3.5) is evaluated per unique source against its reference set, which is why the two sections report different n . TF-IDF with (1, 2)-gram features and class-weighted logistic regression establishes the lexical baseline. DistilBERT, a general-domain contextual model, actually *underperforms* the lexical baseline by 7.5% relative. BioBERT, pretrained on biomedical text specifically, improves over TF-IDF by 9.7% relative, and this result reproduced independently within 1% absolute macro-F1

Model	Macro-F1
TF-IDF + Logistic Regression	0.424
DistilBERT	0.392
BioBERT	0.465

Table 2: Operation classification on the PLABA validation set ($n = 1,431$ pseudo-labeled source-target pairs).

Model	Macro-F1
TF-IDF only	0.424
+ UMLS only (3 feat.)	0.443
+ UMLS + Numerical (6 feat.)	0.445
+ UMLS + Warning + Num. (8 feat.)	0.449

Table 3: Feature-integration ablation, TF-IDF backbone, corrected for the vocabulary-size confound.

across three separate training runs (0.465, 0.461, 0.467), indicating the result is not an artifact of training randomness. We read this pattern as evidence that biomedical-domain pretraining is valuable for operation prediction, since BioBERT outperforms both the lexical baseline and the evaluated general-domain contextual baseline; we stop short of claiming this isolates domain adaptation as the sole cause, since DistilBERT and BioBERT also differ in architecture (6 vs. 12 transformer layers) and parameter count, confounded with pretraining domain in this comparison.

BioBERT’s confusion matrix reveals where difficulty concentrates across all three training runs: Generalization is consistently its weakest class (F1 0.365–0.423, recall 29–32% depending on the run), confused almost equally with both Substitution and Explanation, whereas contextual pretraining substantially improved Explanation recall over the TF-IDF baseline (35% $\rightarrow$ 55% in the run used for the deployed classifier). We attribute Generalization’s persistent difficulty at least partly to the pseudo-label noise described above, rather than purely to architectural capacity, since label noise likely limits achievable classifier performance.

Feature integration. With TF-IDF vocabulary size controlled across conditions, engineered features improve macro-F1 from 0.424 to 0.449 (Table 3). UMLS features account for most of the gain; warning features contribute negligibly to routing accuracy and are instead used downstream as safety constraints (Section 3.4).

Numerical features alone measurably *hurt* Generalization F1 (0.410 $\rightarrow$ 0.387), which we attribute to the numerical signal being entangled with the

same pseudo-label noise affecting Generalization generally, rather than the feature itself being uninformative.

We also integrated engineered features directly with BioBERT. Although validation macro-F1 increased from 0.465 to 0.483, downstream test SARI decreased from 21.72 to 21.08, suggesting that improved operation-classification accuracy under noisy pseudo-labels does not necessarily translate into better end-to-end simplification.

3.4 Operation-Aware Generation and CHV-then-Polish

The predicted operation determines the generation strategy: Substitution routes to CHV lookup, a deterministic dictionary substitution; Explanation and Generalization route to FLAN-T5 with an operation-specific constrained prompt in place of a single generic instruction.

An initial version of the full pipeline, evaluated against the three baselines on the complete validation set, scored a SARI of only 18.76 – slightly below Rule-based CHV’s 19.06 – because 43% of sentences (Substitution-routed) received only a flat dictionary word-swap with no fluency improvement. We addressed this with a second generation step for Substitution-routed sentences: after CHV substitution, an optional FLAN-T5 fluency-polish pass rewrites the sentence for naturalness, constrained by an explicit list of protected spans drawn directly from our complexity detectors – the numerical expressions and the CHV replacement terms just inserted – injected into the prompt as spans that must not change, and *verified post-hoc*: if any protected span is absent from the polished output, the pipeline discards it and falls back to the deterministic CHV-only result rather than trusting the instruction was followed. This is our Detect-Twice, Verify-or-Fallback mechanism (Section 2) instantiated concretely.

An intermediate version of this mechanism, protecting only numerical spans and CHV replacement terms, raised SARI to 20.09 but measurably reduced warning preservation from 1.000 to 0.986: a Substitution-routed sentence that happened to also contain a warning phrase had that phrase left unprotected, and the polish pass occasionally reworded it. Adding the warning detector’s matched phrases to the same protected-span list fully recovered warning preservation to 1.000 *and* pushed SARI higher still, to 20.22 – a genuine improvement on both

System	SARI	Warn.	Num.
No simplification	18.45	1.000	1.000
Rule-based CHV	21.12	1.000	1.000
Direct LLM (FLAN-T5)	25.49	0.980	0.925
Operation-aware	21.72	1.000	0.963

Table 4: Full-system comparison, PLABA *test* set, overall split ($n = 1,000$). “Warn.”/“Num.” are warning/numerical preservation.

axes from the same fix, not a trade-off, and itself a small demonstration of the paper’s broader argument: the failure mode only became visible, and only became fixable, because we evaluated on both a readability metric and a safety metric simultaneously.

Scope of the protected-span guardrail. The protected-span verification (numerical values, warning phrases, and CHV replacement terms alike) currently applies only to Substitution-routed CHV-then-polish outputs; Explanation and Generalization use constrained prompts without equivalent post-hoc verification for either warnings or numbers. On the validation warning subset, only 29/73 sentences were routed through the protected path, yet the full system retained 1.000 warning preservation. We treat this as an empirical observation rather than a guarantee; the overall-split numerical-preservation gap (0.963, Section 3.5) is the more visible cost of this same scope limitation. Extending verification to all generative routes is future work (Section 5).

3.5 Full-System Evaluation

We report the complete comparison on the held-out PLABA *test* set ($n = 1,000$ sentences overall; $n = 49$ warning-bearing sentences in the stratified stress test). All pipeline choices were fixed during validation before evaluation on the held-out test set. The deployed BioBERT checkpoint achieved 0.467 macro-F1 – a separate run from the one whose confusion matrix is discussed in Section 3.3, within the same three-run reproducibility range – and all FLAN-T5 systems used matched beam-search decoding.

On the full test split (Table 4), operation-aware simplification improves over CHV in SARI (21.72 vs. 21.12) and matches its warning preservation (1.000), although numerical preservation falls from 1.000 to 0.963. We attribute this specific gap to the same scope limitation as the warning guardrail (Section 3.4): numerical spans are only pro-

System	SARI	Warn.	Num.
No simplification	18.79	1.000	1.000
Rule-based CHV	22.05	1.000	1.000
Direct LLM (FLAN-T5)	24.73	0.980	0.818
Operation-aware	25.43	1.000	0.909

Table 5: Full-system comparison, PLABA *test* set, warning-stratified stress test ($n = 49$; small n – treat as indicative, see Section 3.1).

tected and verified on the Substitution path, so Explanation- or Generalization-routed sentences containing a number can have it altered during unverified FLAN-T5 generation. Relative to direct FLAN-T5, it trades SARI (21.72 vs. 25.49) for stronger warning (1.000 vs. 0.980) and numerical preservation (0.963 vs. 0.925). Thus, the overall result is a controlled trade-off rather than universal dominance.

On warning-bearing sentences (Table 5), the operation-aware system exceeds direct FLAN-T5 across all three measured dimensions: SARI 25.43 vs. 24.73, warning preservation 1.000 vs. 0.980, and numerical preservation 0.909 vs. 0.818. Because $n = 49$, we treat the magnitude as indicative; importantly, the same qualitative pattern appeared during validation. Together with BioBERT’s stable performance across three runs (Section 3.3), this suggests the result is not driven by one favorable checkpoint or split.

Figure 1 and Figure 2 visualize both tables directly on the readability-safety plane: the overall split shows a genuine trade-off frontier between our system and Direct LLM, while the stratified split shows our system strictly dominating Direct LLM on both axes simultaneously.

4 Related Work

PLABA and prior operation-aware work. PLABA (Attal et al., 2023) pairs biomedical abstracts with expert plain-language rewrites at the sentence level, letting us pseudo-label operations rather than treat simplification as opaque sequence-to-sequence generation. Knappich et al. (2023)’s PLABA-2023-winning BoschAI system finds that fine-tuned models copy the input and edit little, directly motivating our diagnostic framing; Li et al. (2024) investigate BART-based and LaySciFive LLMs with controllable attributes on the same track, though we compare against a general-purpose LLM baseline (FLAN-T5) and a rule-based baseline instead, for reasons discussed in

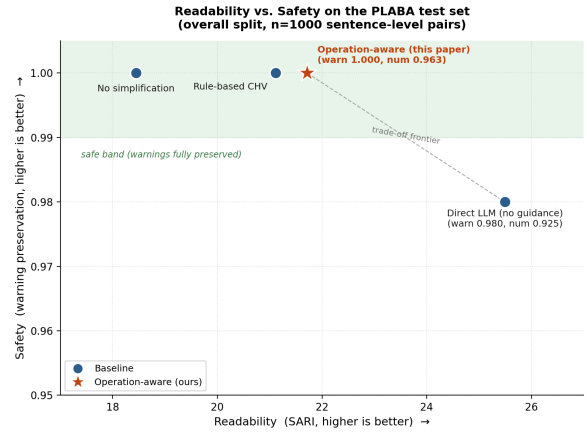


Figure 1: Readability (SARI) vs. safety (warning preservation) on the overall test split ($n = 1,000$). Our system and Rule-based CHV both sit in the full-warning-preservation band; our system reads better. Direct LLM reads better still but exits that band.

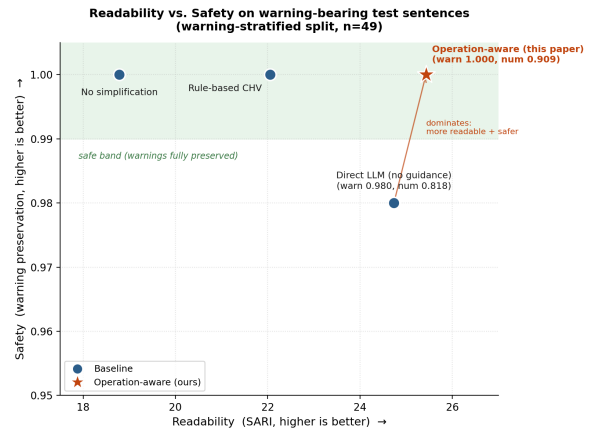


Figure 2: Same axes, restricted to the warning-stratified stress test ($n = 49$). Here our system dominates Direct LLM outright – higher SARI and higher warning preservation simultaneously.

Section 5. Cripwell et al. (2022) propose controllable simplification via operation classification as a predict-then-execute process, the same structural intuition we extend to biomedical text at the sentence level, and Xia et al. (2025) introduce JEBS on the same five-operation PLABA taxonomy, arguing that separating identification, classification, and generation enables more targeted evaluation – the diagnose-before-simplify principle underlying our pipeline.

Jargon detection and safety. Papandreou et al. (2025) show jargon detectors trained on MedReadMe transfer poorly to PLABA (33.71% entity F1), supporting our decision to build detectors natively on PLABA rather than reuse

a mismatched one; their jargon-aware LLM prompts also introduce factual errors a lookup table structurally cannot produce, motivating our CHV-based Substitution pathway. Devaraj et al. (2021) identify safety-critical information loss as a recurring failure in paragraph-level medical simplification, directly motivating our preservation metrics, and Rahman and Lybarger (2025) find that stylistic perturbations degrade LLM QA correctness even when fluency stays stable – external evidence that surface-level metrics can miss real content degradation, the same gap our safety metrics target.

The evaluation gap. SARI (Xu et al., 2016) and BERTScore (Zhang et al., 2020) reward correct edit operations and semantic similarity respectively, but neither captures whether a warning or dosage survived, consistent with reports that SARI and FKGL correlate only moderately with human judgment (Alva-Manchego et al., 2021). This gap holds even for the strongest recent systems: the winning TSAR 2025 submission (Miyata et al., 2025) optimizes candidate diversity and reranking rather than edit-type understanding, and the shared task’s own findings (Alva-Manchego et al., 2025) report that none of 48 submissions classified complexity type before generating output, with organizers explicitly calling for expert-in-the-loop validation as future work. Our preservation metrics answer this call by making safety a first-class, automatically measurable criterion rather than deferring it to expert review.

5 Conclusion and Future Work

We present a sentence-level operation-aware framework for biomedical text simplification that routes Substitution, Explanation, and Generalization through different simplification mechanisms rather than treating all inputs as unrestricted generation. Under weak supervision, BioBERT provides the strongest evaluated routing model, while UMLS-based features add signal beyond TF-IDF lexical features. End-to-end evaluation shows that this architecture moves beyond deterministic lexical substitution while improving warning and numerical preservation relative to direct generation, via our Detect-Twice, Verify-or-Fallback mechanism. On warning-bearing test sentences, it also exceeds direct FLAN-T5 in SARI and both preservation metrics, although the small sample warrants cautious interpretation.

Our evaluation also highlights a methodological issue: aggregate or random-sample evaluation can hide rare safety-critical failures. Warning-stratified testing exposed a failure not observed under random sampling, suggesting that safety-sensitive NLP evaluation should pair overall quality metrics with targeted stress tests on high-risk subsets.

Future work should replace weak sentence-level pseudo-labels with gold or span-level operation annotations, extend protected-span verification to Explanation and Generalization, evaluate preservation with human judgments, and compare against specialized biomedical simplification models such as the BART-based and Lay-SciFive systems of Li et al. (2024).

Limitations

Our operation labels are weak sentence-level pseudo-labels and cannot represent multiple operations within one sentence. UMLS jargon detection is evaluated primarily through abstract-level hit rates and retains phrase-level false positives. Protected-span verification currently covers only the Substitution route, while Explanation and Generalization remain prompt-constrained but unverified. Preservation metrics rely partly on lexical matching and do not fully recognize valid paraphrases. The warning-stratified test contains only 49 sentences, and evaluation is limited to English PLABA data; broader datasets and human safety judgments are needed.

Ethics Statement

This work addresses a patient-safety concern: automated biomedical text simplification that silently drops warnings or dosages could cause harm if deployed without safeguards. None of the systems described here, including our own, are validated for unsupervised clinical deployment; any real-world use should retain human oversight (e.g., pharmacist or clinician review) before reaching patients. We used only publicly available, de-identified research abstracts (PLABA), and all models used (BioBERT, DistilBERT, FLAN-T5) are open-weight and run locally, so no data processed by this pipeline is transmitted to a third-party API.

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